Analyst Forecast Bias

Structure, Regime Sensitivity, and Predictive Signals

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Stack: Python | PyMC 5.28 | ArViZ 0.23 | XGBoost 3.2 | hmmlearn | pandas | yfinance | matplotlib

Wall Street analysts are systematically wrong about earnings. They underestimate more often than they overestimate, and they underestimate by different amounts for different types of companies. This project asks three questions about that bias: does it vary by the type of company being forecast, does the macro environment change how large it is, and can the pattern of analyst revisions in the weeks before an announcement predict whether a beat or miss is coming?

Data

The primary dataset is a panel of quarterly EPS actuals and consensus estimates for 72 large-cap US companies from Q4 2017 through Q2 2026. After cleaning and winsorizing forecast errors at the 2.5th and 97.5th percentiles within each category, the panel contains 2,455 company-quarter observations, balanced across categories at roughly 406 to 415 rows each.

A second dataset of 63,744 weekly consensus snapshots captures how analyst estimates evolve in the 16 weeks before each announcement. This is the raw material for the revision path features.

Macro data - quarterly S&P 500 returns and quarterly average VIX - is downloaded via yfinance and merged onto each company-quarter by date. Both are standardized to z-scores before entering the model so their coefficients are directly comparable.

Category Definitions

Companies are grouped into six earnings-driver categories. Defensive baseline (12 tickers: JNJ, PG, KO and similar) serves as the control group - stable, predictable earnings, deep analyst coverage. The five other categories are:

â€¢ Commodity driven (XOM, CVX, FCX and similar) - earnings tied to commodity prices

â€¢ Macro rate sensitive (JPM, GS, BAC and similar) - driven by interest rate spreads and credit conditions

â€¢ Tech cycle (NVDA, AAPL, ASML and similar) - driven by semiconductor and product cycles

â€¢ Economically cyclical (CAT, F, AMZN and similar) - tracks the broader economic cycle

â€¢ Regulatory idiosyncratic (LLY, PFE, NEE and similar) - earnings shaped by regulatory events and approvals

If the bias patterns found in the other five categories are real and not artifacts of the data or methodology, they should not appear in defensive baseline. This is the falsifiability check built into the design.

Methodology

Regime Detection

A three-state Gaussian Hidden Markov Model is fitted to monthly S&P 500 returns from January 2016 to May 2026. The model identifies three latent market states - bull, sideways, and bear - by learning the characteristic return and volatility of each state from the data rather than applying manual labels.

The model converged cleanly. The three states separate primarily on volatility rather than return direction. The bear state has a mean monthly return of 0.002 and a standard deviation of 0.075 - nearly identical in return to the sideways state but dramatically more volatile. This is economically sensible: the defining feature of a crisis regime is not that returns are negative but that uncertainty is extreme. March and April 2020 are correctly labelled bear; June 2021 bull; June 2022 bear.

These regime labels informed the EDA but were ultimately replaced by continuous VIX in the Bayesian model. With only nine years of data and three to four years per regime, the model could not estimate separate regime means - the posterior intervals were too wide to be useful. Continuous VIX gives every observation its own macro context and produces estimable coefficients.

Hierarchical Bayesian Model

The core model is a three-level hierarchical Bayesian regression fitted in PyMC. The target is winsorized forecast error at the company-quarter level. The hierarchy has three levels: a grand mean shared across all observations, category-level offsets that capture persistent bias by earnings-driver type, and continuous macro predictors (z-scored VIX and S&P 500 return) that capture how the macro environment modulates bias.

The likelihood is Student-t rather than Normal. QQ plots for all six categories show consistent right skew and heavy tails, with normality correlations ranging from 0.707 to 0.931. The model confirms the choice - the posterior for the degrees of freedom parameter nu concentrates tightly around 1.15 (94% HDI [1.056, 1.232]), far below the threshold of roughly 30 where Student-t approximates Normal. Using a Normal likelihood would systematically understate the probability of large earnings misses.

The model is sampled with four NUTS chains, 1,000 tuning steps, and 2,000 draws per chain. Convergence is clean: R-hat = 1.000 across all parameters, minimum effective sample size = 1,396, zero divergences.

Revision Path Features

Seven features are constructed from the weekly consensus snapshots for each company-quarter. All are observable before the earnings announcement and contain no forward-looking information:

â€¢ Revision slope - linear trend of consensus estimates over the 16-week window

â€¢ Revision acceleration - whether the revision trend is gaining or losing momentum

â€¢ Direction changes - how many times consensus reversed direction

â€¢ Final vs initial - net change in consensus from eight weeks before announcement to the day before

â€¢ Spread trend - whether analyst disagreement (high minus low estimate) is growing or shrinking

â€¢ Analyst count trend - whether coverage is growing or shrinking into the quarter

â€¢ Weeks of data - number of weekly snapshots available

XGBoost Classifier

An XGBoost binary classifier is trained to predict beat or miss from the seven revision path features plus category (integer encoded), z-scored VIX, and z-scored S&P 500 return. The train/test split is strictly time-based: 2017 to 2022 for training, 2023 to 2025 for testing. A random split is not used because it would allow future quarters to leak into the training set.

The 82% beat rate creates a class imbalance problem. Accuracy is a useless metric here - a naive always-beat classifier scores 82%. The model uses scale_pos_weight to upweight misses during training and is evaluated on AUC-ROC and precision-recall AUC instead.

Findings

1. Market uncertainty amplifies analyst error - market direction does not

The VIX coefficient has a posterior mean of 0.006 EPS and a 94% credible interval of [0.001, 0.010] - it excludes zero. A one-standard-deviation increase in quarterly VIX is associated with a credible increase in analyst underestimation. The S&P 500 return coefficient has mean 0.004 and interval [-0.001, 0.008] - it includes zero and shows no credible effect.

Analysts miss more in high-uncertainty quarters, not simply in down quarters. A volatile but flat quarter is more disruptive to forecasting than a steadily declining one. This is a more precise claim than the usual narrative that bear markets cause more analyst misses - the data points to uncertainty, not direction, as the mechanism.

2. Bias varies by earnings-driver category, and the control group holds

Bootstrap confidence intervals on mean forecast error (10,000 resamples) show positive bias in every category, with all intervals excluding zero. The means range from 0.054 EPS for defensive baseline to 0.249 for macro rate sensitive. Tech companies are consistently underestimated regardless of year. Defensive companies are underestimated least.

In the Bayesian model, defensive baseline is the only category whose posterior credible interval excludes zero (mean -0.034, HDI [-0.061, -0.010]). The control group holds - stable earnings and deep analyst coverage produce the lowest and most credible bias estimate in the dataset.

The category by year heatmap surfaces one result worth highlighting separately: economically cyclical in 2022 shows a mean forecast error of -0.15, the only negative cell in the entire dataset. Analysts overestimated consumer and industrial earnings in the one year a rapid rate shock hit spending hard and fast. The sign of the bias flipped - and it flipped for exactly the category you would expect it to flip for.

3. Revision path signal is real but category-dependent

The XGBoost classifier achieves AUC-ROC of 0.573 on the held-out 2023-2025 test set against a random baseline of 0.500. PR-AUC is 0.900 against an always-beat baseline of 0.860. The lift is modest but consistent across all probability thresholds, and the calibration plot confirms the model's output is a reliable ordinal ranking - higher predicted beat probability corresponds monotonically to higher actual beat rate across all bins.

The per-category AUC breakdown is the most practically useful result:

â€¢ Tech cycle: 0.673 - revision path is genuinely predictive. Analysts revise upward before tech announcements but systematically undershoot, and this pattern is learnable

â€¢ Defensive baseline: 0.567 **|** Economically cyclical: 0.551 **|** Macro rate sensitive: 0.537 - all above random

â€¢ Commodity driven: 0.440 **|** Regulatory idiosyncratic: 0.453 - both below random

For commodity and regulatory companies the revision path actively misleads. Commodity earnings are driven by oil and metals prices that move after analysts have submitted their estimates. Regulatory earnings turn on binary FDA approval decisions. No amount of watching consensus revisions helps when the outcome-determining event has not yet happened.

Feature importance (gain-based): category membership is the single strongest predictor at 14.6% of total gain. The revision path features collectively contribute 56%, with the net consensus change in the final eight weeks (12.2%) and revision slope (9.9%) as the strongest individual contributors. VIX and S&P 500 return together add 17%. All three layers of the project show up in the feature ranking in the right order.

Limitations

72 companies over nine years is enough to find patterns but limits precision. The regime effect cannot be estimated discretely with this sample - replacing bull/bear/sideways labels with continuous VIX was a direct consequence of having only three to four years per regime. A dataset with 20-plus years and 300-plus companies would allow more granular analysis.

Survivorship bias is present. The 72 companies are current large-cap constituents - chronic missers that shrank or were acquired over the period are not included, which likely inflates beat rates slightly. Category assignments are also fixed; in reality a company's earnings drivers shift over time.

The classifier is tested on three years of held-out data, and 2023-2025 has structural features that differ from the training period - AI-driven tech outperformance, higher rates. The per-category AUC results should be treated as directional rather than precise estimates.

Full code, notebooks, and figures at github.com/souro26/analyst-forecast-bias